Week 30: Two-pan weight kits

Adult guide built from the problem files · Unpiloted. Physical pretests unperformed.

This week's adult guide is week-30-facilitator.pdf, whose source is editable/facilitator-src/guide.json. These problem files hold each problem's one-sentence idea, not its answers; the answers, hints and the counting bound are in the guide.

Problems and routes

Problem	K–1	2–3	4–5	route
Balance targets 1 to 4 with weights 1 and 3	P1			first
Which targets weights 1 and 2 balance	P2			first
Kit 1 and 4 against kit 1 and 3	P3			first
Every two-weight kit for targets 1 to 4	P4		P1	K-1 return; 4-5 first
Kits 1 and 2, 1 and 3, 1 and 4 on targets 1 to 6		P1		first
A third weight for 1 and 3	P5	P2	P2	K-1 return; 2-3 first; 4-5 first
The first target 1, 3, 8; 1, 3, 9; 1, 3, 10 cannot balance		P3		first
Every way to balance 4 with 1, 3, 8 and with 1, 3, 9		P4		return
Can 1, 3 and 9 balance a target two ways?			P3	return
Can three weights balance 1 to 14?		P5	P4	return
How many targets can three weights balance?			P5	return
A fourth weight for 1, 3 and 9		P6	P6	return
The longest run with five weights			P7	return

First route. K–1 P1–P3; Grades 2–3 P1–P3; Grades 4–5 P1–P2. **Return visits.** K–1 P4–P5; Grades 2–3 P4–P6; Grades 4–5 P3–P7.

Key, one entry per problem

Balance targets 1 to 4 with weights 1 and 3

K–1 P1 · first route

Idea. Weights 1 and 3 balance 1, 3 and 4 directly and 2 by putting 1 beside the target: $2 + 1 = 3$.

Which targets weights 1 and 2 balance

K–1 P2 · first route

Idea. Weights 1 and 2 balance targets 1, 2 and 3 only; 4 and 5 are more than both together.

Kit 1 and 4 against kit 1 and 3

K–1 P3 · first route

Idea. Kit 1 and 4 balances 1, 3, 4 and 5 (missing 2 and 6); kit 1 and 3 balances 1 to 4 with no gap, so the spread-out kit is not the better one.

Every two-weight kit for targets 1 to 4

K–1 P4 · Grades 4–5 P1 · K-1 return; 4-5 first route · wording differs by band

Idea. Only 1 and 3 balances every target from 1 to 4: two weights give at most four positive targets, so there can be no repeat and no gap.

Kits 1 and 2, 1 and 3, 1 and 4 on targets 1 to 6

Grades 2–3 P1 · first route

Idea. Kit 1 and 2 reaches 1 to 3, kit 1 and 3 reaches 1 to 4, and kit 1 and 4 reaches 1, 3, 4, 5 with a gap at 2.

A third weight for 1 and 3

K–1 P5 · Grades 2–3 P2 · Grades 4–5 P2 · K-1 return; 2-3 first; 4-5 first route · wording differs by band

Idea. Adding 9 to 1 and 3 balances every target from 1 to 13: 8 leaves a repeat and stops at 12, 10 leaves a gap at 5.

The first target 1, 3, 8; 1, 3, 9; 1, 3, 10 cannot balance

Grades 2–3 P3 · first route

Idea. 1, 3, 8 first fails at 13, 1, 3, 9 at 14, and 1, 3, 10 at 5.

Every way to balance 4 with 1, 3, 8 and with 1, 3, 9

Grades 2–3 P4 · return route

Idea. With 1, 3, 8 target 4 balances two ways (1 + 3, and 8 against 4 + 1 + 3); with 1, 3, 9 only one.

Can 1, 3 and 9 balance a target two ways?

Grades 4–5 P3 · return route

Idea. No: 1, 3, 9 has 27 placements, the all-off one and 13 pairs of opposites, so the 13 positive targets each get exactly one.

Can three weights balance 1 to 14?

Grades 2–3 P5 · Grades 4–5 P4 · return route · wording differs by band

Idea. No three-weight kit balances 14 different positive targets: its 27 placements give at most 13.

How many targets can three weights balance?

Grades 4–5 P5 · return route

Idea. Any three weights balance at most 13 different positive targets, gaps or not; 1, 3, 9 reaches 13.

A fourth weight for 1, 3 and 9

Grades 2–3 P6 · Grades 4–5 P6 · return route · wording differs by band

Idea. Adding 27 to 1, 3, 9 balances every target from 1 to 40, each in exactly one way: the new weight is twice the old total plus 1.

The longest run with five weights

Grades 4–5 P7 · return route

Idea. Five weights balance at most $(3^5 - 1)/2 = 121$ positive targets; 1, 3, 9, 27, 81 reaches every one from 1 to 121.